MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8167

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, metal-ceramic

See figure 1

Mounting position: Any

Weight: 4 ounces (105.4 grams) nominal

ABSOLUTE RATINGS: F = 500 MHz

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ehk	Pg1	Pg2
Unit:	Vac (Note 1)	Vdc	Vdc	Vdc	mAdc	Vdc	W	W
Maximum:								
C Telep:	6.0 ±5%	1,500	-250	300	200	±150	2	12
C Teleg:	6.0 ±5%	2,000	-250	300	250	±150	2	12
Class AB1:	6.0 ±5%	2,500	---	400	250	±150	2	12
TEST CONDITIONS:	6.0	1,000	Adj	300	150	---	---	---

ABSOLUTE RATINGS: F = 500 MHz

Parameter:	Pp	T (seal and anode core)	tk	Cooling	Barometric pressure, reduced
Unit:	W	°C	sec (min)	Note 2	mmHg (min)
Maximum:					
C Telep:	200	250	30	---	141 ±5
C Teleg:	300	250	30	---	141 ±5
Class AB1:	300	250	30	---	141 ±5
TEST CONDITIONS:	---	---	120	Note 3	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1256	Electrode current (screen)	-		0.65	II	Ic2	---	-5.0 +3.0	mAdc
1266	Total grid current	4	Eb = 2,000 Vdc	0.65	II	Ic1	---	-15	µAdc
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-32.0	-45.0	Vdc
1266	Primary grid emission (control)	-	Ic1 = 70 mAdc; t = 15 seconds; anode and g2 floating	0.65	II	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; t = 15 seconds; Ic2 = 100 mAdc; anode floating	0.65	II	Isg2	---	-250	µAdc
(D) 2212	Pulse emission (1)	5	Eb = Ec2 = 250 Vdc; Ec1= -100 Vdc; egk/ik= 1.5a; Ef = 5.4 Vac; tp = 4,500 µs (min); prr = 11 ±1; tr = tf = 25 µs; slope = 0.5%; ripple = 0.1%	0.65	II	Δik	---	200	ma
1372	Current division (method A)	-	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/Ib = 1.0a; tp = 4,500 µs (min); prr = 11 ±1	0.65	II	egk ic1 ic2	8.0 --- ---	18.0 250 250	v ma ma
	Quality conformance inspection, part 2								
1301	Heater current	-		---	---	If	2.6	3.1	Aac
---	Stability	-	Eb = 2,000 Vdc; Ec1/Ib = 150 mAdc at t = 0; read ΔIb at t = 120 seconds	---	---	ΔIb	---	10	mAdc
1366	Interelement leakage resistance, cold	-	Rs = 2.5 Meg E = 100 Vdc, g1 negative E = 500 Vdc, g2 positive E = 500 Vdc, p positive	---	---	Rg1k Rg1g2 Rg2p	50 50 50	--- --- ---	Meg Meg Meg
1331	Direct-interelectrode capacitance (ground cathode connection)	-	Test in fixture per DESC Drawing 67001, or equivalent	---	---	Cgp Cin Cout	--- 25.0 3.5	0.06 33.0 4.5	pF pF pF
1336	Heater-cathode leakage	-	Ehk = +250 Vdc Ehk = -250 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
(D) 2212	Pulse emission (2)	5	Pulse emission (1), except Ef = 6.0 Vac	---	---	Δik	---	100	ma
(D) 2214	RF useful power output	-	Class C amplifier; F = 450 to 500 MHz; Eb = 2,000 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 5.0 Vac; circuit and cavity per Drawing 285-JAN, or equivalent	---	---	Po	225	---	W

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Vibration (noise)	6,7	Ef = 6.0 V (ac or dc); Ebb = 2,500 Vdc; Ec2 = 350 Vdc; Rp = 4,900 ohms; Ec1/Ib = 100 mAdc; accel = 20 G peak (min); F = 100 to 750 Hz, ascending sweep only	---	---	Ep	---	20	Vac
---	Vibration (noise) end points:								
1266	Total grid current	-		---	---	Ic1	---	-20	µAdc
1261	Electrode voltage (grid)	-		---	---	Ec1	-32.0	-45.0	Vdc
---	Long duration shock	6,8	Eb = 2,000 Vdc; Ec2 = 350 Vdc; Ec1 = -150 Vdc; shock = 11 ±2 ms; accel = 90 G peak (min); total impacts = 18	---	---	---	---	---	---
---	Long duration shock end points:								
1266	Total grid current	-		---	---	Ic1	---	-20	µAdc
1261	Electrode voltage (grid)	-		---	---	Ec1	-32.0	-45.0	Vdc
(D) 2204	Linear amplifier power output and distortion	6	Eb = 2,000 Vdc; Ec2 = 350 Vdc; Ec1/Ibo = 125 mAdc; Eg1/Ib = 250 mAdc; 1 tone; Rl = 4,150 ±150 ohms; Rg = 1,000 ohms (max); F = 2 to 10 MHz; anode tank Q = 10 to 15	---		Po	225	---	W (useful)
						3rd IM	-25	---	dB
						5th IM	-30	---	dB
(D) ---	Life test (5)	12	Group C; linear amplifier power output and distortion; t = 500 hours	---	---	Po (initial)	225	---	W (useful)
---	Life-test (5) end points:								
1366	Interelement leakage resistance, cold	15		---	---	Rg1k	10	---	Meg
						Rg1g2	10	---	Meg
						Rg2p	15	---	Meg
(D) 2204	Linear amplifier power output and distortion	-		---	---	Po	210	---	W (useful)
						3rd IM	-24	---	dB
						5th IM	-29	---	dB
(D)	Life test (5)	12	t = 500 hours	---	---	Po	210	---	W (useful)
						ΔIb	---	35	mAdc
---	Life test (2)	-	Group C; Ef = 6.6 Vac; Ec1 = Ec2 = Eb = 0; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test (2) end points:								
1366	Interelement leakage resistance, cold	15		---	---	Rg1k Rg1g2 Rg2p	10 10 15	--- --- ---	Meg Meg Meg
---	Life test (4)	13	Group C; Ec1 = -100 Vdc; g2 and p floating; t = 200 hours	---	---	---	---	---	---
---	Life-test (4) end points:								
1301	Heater current	-		---	---	If	2.6	3.1	Aac
1336	Heater-cathode leakage	-	Ehk = +250 Vdc	---	---	Ihk	---	150	µAdc
			Ehk = -250 Vdc			Ihk	---	150	µAdc
(D) 1155	Pressure drop	11	Airflow = 5.0 cfm (min)	---	---	---	---	0.4	In. H_2O
---	Cooling	11,14	Eb = 2,000 Vdc; Ec1 = 0; Ec2/Ib = 150 mAdc; airflow = 5.0 cfm (max)	---	---	T(anode) core	---	250	°C
						T(anode seal)	---	225	°C
						T(base seal)	---	225	°C
(D) 1011	Humidity	9,11		---	---	---	---	---	---
---	Humidity test end point:								
1266	Total grid current	10		---	---	Ic1	---	-20	µAdc
---	Life test (1)	11	RF useful power output; t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end points:								
2212	Pulse emission (2)	-		---	---	Δik	---	100	ma
1266	Primary grid emission (control)	-		---	---	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-250	µAdc
1336	Heater-cathode leakage	-	Ehk = +250 Vdc	---	---	Ihk	---	150	µAdc
			Ehk = -250 Vdc			Ihk	---	150	µAdc
2214	RF useful power output	-		---	---	Po	180	---	W
---	Life test (3)	11	Ebb = 1,000 Vdc; Ec2 = 250 Vdc; Ec1/Ib = 50 mAdc; Eg1/Ib = 100 mAdc; Rp = 1,000 ohms; TA = 200°C (min); F = 1 kHz (min); t = 100 hours	---	---	---	---	---	---
---	Life-test (3) end points:								
2212	Pulse emission (2)	-		---	---	Δik	---	100	ma
1266	Total grid current	-		---	---	Ic1	---	-20	µAdc

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NOTES:

1. At frequencies above approximately 200 MHz, it may be necessary to reduce heater voltage to compensate for rf transit-time heating of the cathode after dynamic operation of the tube has started. This back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. Particular care should be used in the selection of stable circuit components and in final tuning of high-frequency circuits as off-resonance operation, even to a small degree, may result in marked and undesirable increase in cathode temperature. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for any particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. For straight-through, Class C CW amplifier operation, the following heater operation voltages are indicated:

Frequency (MHz)	Ef (Vac)
201 to 300	5.75
301 to 400	5.50
401 to 500	5.00

2. When the tube is operated at 100 percent of maximum rated anode dissipation at an incoming air temperature of 25°C maximum, a minimum airflow of 5.0 cfm at sea level shall pass through the anode cooler. If the socket EIMAC SK-710 and chimney SK-606, or equivalents, are used, an incoming airflow of 5.0 cfm to the lower end of the socket is required. At this flow of 5.0 cfm, the static pressure drop directly across the tube and socket (with chimney installed) is approximately 0.4 inch of water. This pressure drop varies with the amount of escaping air and with the shape and construction of the air director (chimney). The airflow rating applies at bias voltages of less than 100 volts and frequencies less than 500 MHz. Air cooling of the tube should be increased with increased negative grid bias, increased incoming air temperature, increased frequency of operation, or a combination. In all cases of operation, a socket which provides forced-air cooling of the base shall be used, and maximum seal and anode core temperature ratings shall not be exceeded. Airflow must be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

3. In all electrical tests involving application of heater voltage, forced-air cooling of the tube is allowed at the rate of 5.0 cfm for the base and anode unless otherwise specified herein. The tube should be operated in an air-system socket (EIMAC SK-710 socket, with EIMAC SK-606 chimney, or equivalents. Standard temperature and pressure conditions shall apply.

4. This test is to be the first test performed at the conclusion of the holding period.

5. Pulse emission (1) and (2) are taken at the voltage conditions specified. For the basic test circuit, see test method 1372, figure 1372-1.

Ⓓ 6. Testing shall be performed every three months, with sampling as follows:

n1 = 4	c1 = 0	where c2 represents the total allowable
n2 = 4	c2 = 1;	failures for the first and second samples combined.

The listed tests shall be considered as nondestructive except in case of failure. In case of a sample failure, that test shall become quality conformance inspection, part 2, AQL 6.5, inspection level S3, for three consecutive successful submissions, at which time the test may revert to the quarterly basis.

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NOTES: - Continued

7. The fixture described in Drawing 284-JAN shall be used to hold the tube in the manner indicated. The test circuit shown on figure 3 shall be used, but tubes found to electrically oscillate for causes other than vibration shall not be tested nor rejected on this test. Each tube under test shall be subjected to one sweep cycle in each of the three axes X, Y, and Z. One sweep cycle (100 to 750 Hz, ascending only) shall be covered in 6 to 12 minutes. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration output voltage in each of the three positions. If at the end of 60 seconds the vibration output is increasing, the vibration shall be continued until there is no further increase for 60 seconds. The tubes shall not show noise voltage output in excess of the maximum limit specified, except one intermittent short per tube shall be allowable during this test. The tube shall show no intermittent or tap shorts after the test. In addition to reading noise voltage output with a VTVM, such as an HP400D or equivalent, a permanent recording shall be made using a good quality recorder to produce a plot of noise voltage versus frequency. Noise voltage amplifiers used with the recorder shall have a ±1 dB frequency response over the range to be measured (100 to 750 Hz) and the overall recording equipment shall be capable of fast response in order to show sharp noise voltage spikes resulting from internal tube resonances or other phenomenon. Prominent noise peaks indicated on the recording shall be individually investigated by fixed-frequency operation, and the 60-second operation shall be made at the frequency of highest noise as so selected. The frequency at the extremes of the sweep shall be read with an accuracy of ±1 percent.

8. The fixture described in Drawing 284-JAN shall be used to hold the tube in the manner indicated. The tubes shall be subjected to the specified acceleration in the X, Y, and Z axes, with six shocks in each axis. Tubes showing any permanent shorts, or more than one temporary short during the test, shall be rejected. The applied shock shall be an approximate half-sine wave motion with duration measured at the zero-axis level.

9. The sample of tubes under test shall be subjected to an atmosphere of 95 to 100 percent relative humidity at a temperature of 95° to 100°C for a period of 14 days. A sample of tubes shall be tested monthly. The test shall be considered as nondestructive except in case of failure. Sampling shall be in accordance with MIL-STD-105, inspection level S3. Single-sampling tables shall be used for normal and tightened sampling, and reduced sampling. Sample size shall be based on the estimated production lot size, except that only codes F through Q shall be used for normal or reduced sampling, while only codes G through Q shall be used for tightened sampling; in no case shall the actual monthly submission exceed the estimate upon which the sample size was based. Allowable failures shall be in accordance with 6.5 AQL.

10. Upon satisfactory compliance to this plan for three successive months of production, lots may be prereleased upon the following conditions:
There shall be no defects at the end of the first 7 days; however, the sample shall continue to 14 days or sample failure (7-day failure criteria shall be I_{c1} = -15 µAdc maximum). In the event the sample fails the 14-day criteria, normal inspection without prerelease consideration must be resumed until the next three lots comply with the specification requirements. In the event of failure of a prerelease sample, the manufacturer shall immediately notify the cognizant government inspector, and contracting officer, of the failure.

(D)11. This test shall be performed during the initial production and then annually thereafter. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The annual double sampling plan shall be reinstated after three consecutive samples have been accepted.

12. During the performance of life-test (5), the rf grid driving voltage and the bias voltage shall be monitored and held constant. The bias voltage shall be adjusted to produce the specified value of Ibo and the rf drive shall be adjusted to produce the specified anode current of 250 mAdc during an initial adjustment period not to exceed 4 hours. In no case shall the grid be driven positive with respect to the cathode as indicated by a grid current of 50 µAdc maximum. At the conclusion of the life-test period, the change (ΔIb) in the anode current from the initial value of 250 mAdc, and the power output (Po), both being read in the life-test equipment itself, shall be read and the tube shall meet the listed requirements for these parameters. Prior to the performance of life-test (5), the tube shall have met the requirements of the linear amplifier power output and distortion test.

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NOTES: -Continued

13. Heater voltage shall be cycled approximately 2.0 minutes "on" and 4.0 minutes "off". No grid-cathode shorts are permitted during or after life test. For qualification, data on a sample of 10 tubes shall be submitted.

14. The tube shall be mounted in an air-system socket (EIMAC SK-710, with SK-606 chimney, or equivalents) arranged so that the entire airflow passes through the anode cooler, with air flowing in a base-to-anode direction. Air under standard conditions of temperature and pressure shall be used, or appropriate corrections applied. Temperatures shall be measured by means of thermocouples, located as follows:
Anode core. The thermocouple shall be embedded in the top of the anode core, adjacent to the cooler, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole so as to hold the thermocouple firmly in place.
Anode seal. The thermocouple shall be attached, by any suitable means, to the base of the anode in the immediate vicinity of the junction of the screen-grid/anode insulating dielectric material with the anode.
Base seal. The thermocouple shall be attached, by any suitable means, to the screen-grid contact where it joints the insulating dielectric material.
In all cases, good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated before the cooling test is performed.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

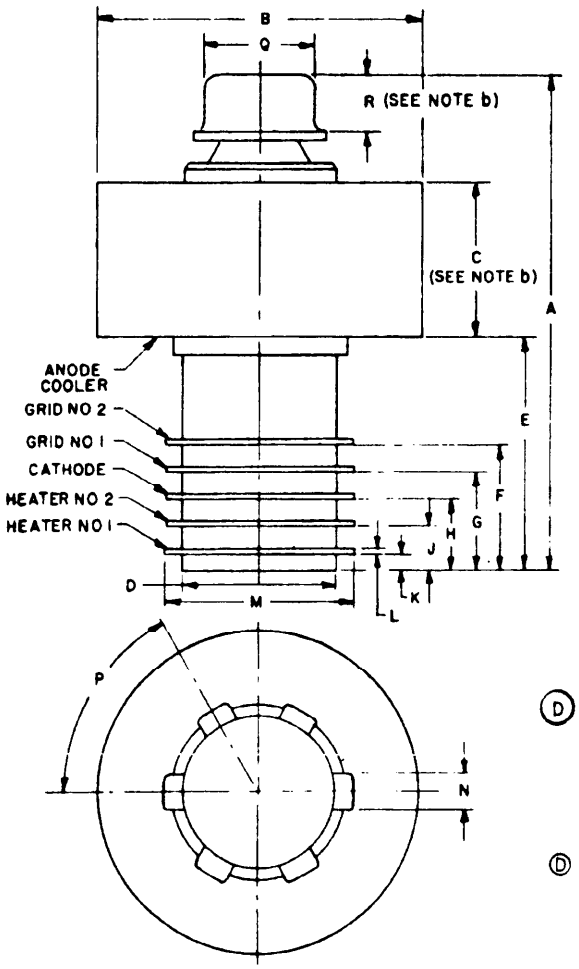
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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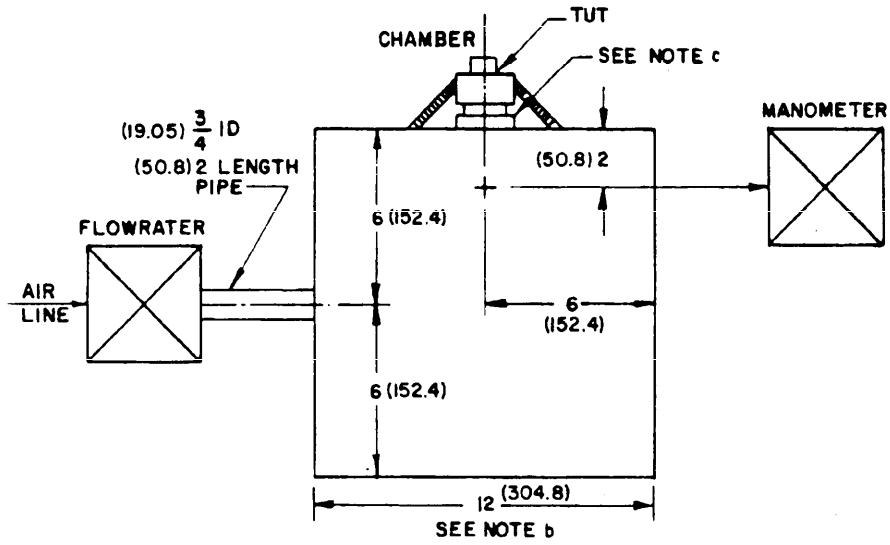


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.300 (58.42)	2.500 (63.50)
D	.740 (18.80)	.770 (19.56)
F	.602 (15.29)	.642 (16.31)
G	.470 (11.94)	.500 (12.70)
H	.329 (8.36)	.359 (9.12)
J	.193 (4.90)	.213 (5.41)
K	.050 (1.27)	.072 (1.83)
L	.010 (.25)	.020 (.51)
M	.936 (23.77)	.956 (24.28)
Quality conformance inspection, part 3 (SEE NOTE a)		
B	1.610 (40.89)	1.640 (41.66)
C	.710 (18.03)	.790 (20.07)
E	1.133 (28.78)	1.195 (30.35)
N	.170 (4.32)	.185 (4.70)
Q	.559 (14.20)	.573 (14.55)
R	.240 (6.10)	
Reference dimension		
P	60°	

NOTES:
a. Dimensions shall be checked during the initial production and annually thereafter. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. Annual double sampling plan shall be reinstated after three consecutive samples have been accepted.
b. Available anode contact surfaces.

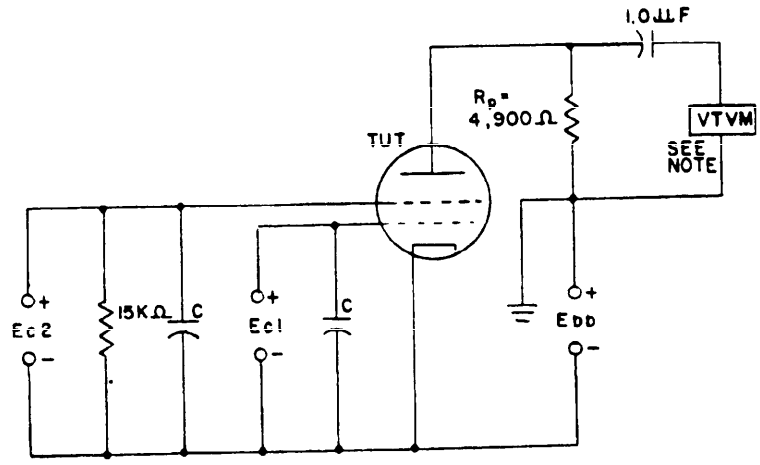
FIGURE 1. Outline drawing of electron tube type 8167.

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NOTES:
a. All dimensions in inches with metric equivalents (mm) in parentheses.
b. 12(304.8) inch cube inside dimension shall be free of air leaks.
c. Socket shall be Eimac No. SK-710, or equivalent.

FIGURE 2. Baffle system.



NOTE: Hewlett-Packard Type 400-D, or equivalent

FIGURE 3. Vibration noise test circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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